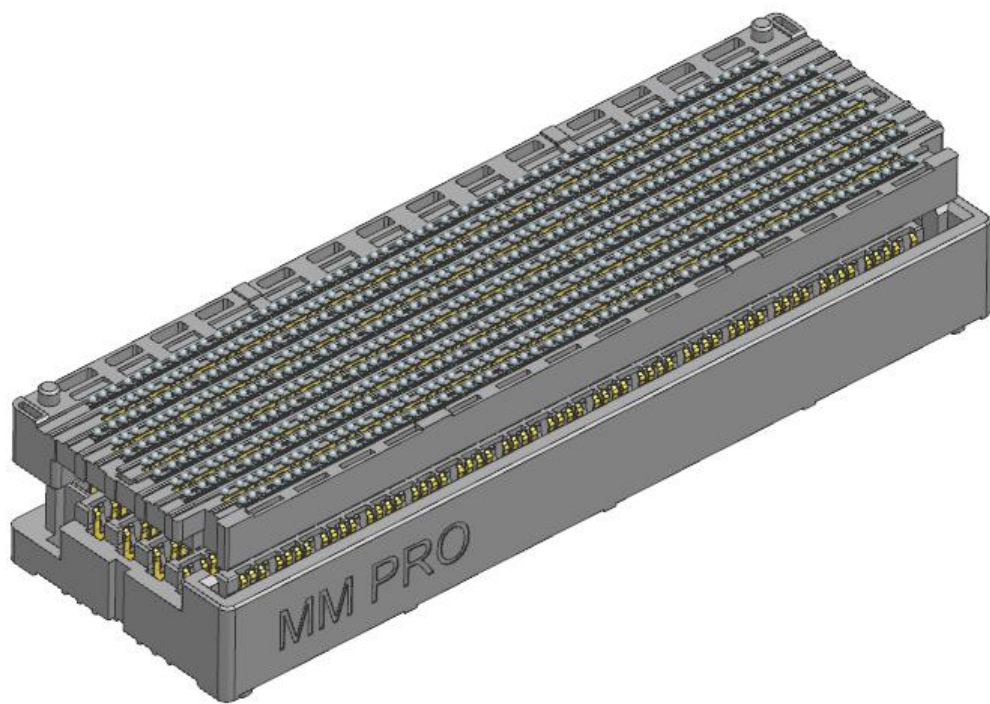


PRODUCT SPECIFICATION FOR
MIRROR MEZZ PRO™



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CHANGE NO.	716999						
REVISED BY	HCGOH	DATE	2022/08/08	DOC TYPE	DOC TYPE DESCRIPTION	DOC PART	SERIES
REV APPR BY	SHONG	DATE	2022/08/21	PS	ENGINEERING SPECIFICATION WORD	000	218910
INITIAL RELEASE				CUSTOMER	DOCUMENT NUMBER	REVISION	SHEET
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1. SCOPE

This Product Specification covers the performance requirement and testing methods for Mirror Mezz Pro™ connectors.

2. PRODUCT DESCRIPTION**2.1 PRODUCT NAME AND SERIES NUMBER(S)**

MIRROR MEZZ PRO™:
218910 218916

2.2 SAFETY AGENCY APPROVALS

UL FILE : TBA
CSA : TBA

3. APPLICABLE DOCUMENTS AND SPECIFICATIONS

EIA-364-1000

4. RATINGS**4.1 VOLTAGE**

<30 Volts AC (RMS) / VDC

4.2 CURRENT

0.75A per pin for 1 oz. Cu Trace
1.2A per pin for 2 oz. Cu Trace

4.3 OPERATING TEMPERATURE

- 55°C to + 105°C

4.4 CYCLE DURABILITY

100 Cycles

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5. PERFORMANCE

5.1 ELECTRICAL REQUIREMENTS

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT
1	Contact Resistance (Low Level)	Mate connectors: apply a maximum voltage of 20 mV and a current of 100 mA. Per EIA-364-23.	30 milliohms MAXIMUM [initial]
2	Insulation Resistance	Un-mate & unmount connectors: apply a voltage of 500VDC between adjacent terminals and between terminals to ground. Per EIA-364-21.	1000 Megohms MINIMUM Signal/Signal
3	Dielectric Withstanding Voltage	Mated connectors: apply a voltage of 500 VAC (RMS) for 1 minute between adjacent terminals. Per EIA-364-20.	No breakdown; current leakage < 5 mA
4	Signal Continuity	Mate connectors. Per EIA-364-87.	No interrupts > 10 nanoseconds
5	Temperature Rise	Mate connectors. Apply a current of 0.75A for 1 oz. Cu trace to 6 adjacent terminals (GGSSGG) on the outer row and measure temperature rise of 2 middle (SS) terminals. Mate connectors. Apply a current of 1.2A for 2 oz. Cu Trace and measure temperature rise.	Temperature rise: 30°C MAXIMUM

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5.2 MECHANICAL REQUIREMENTS

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT
6	Connector Mate and Un-mate Forces	Mate and un-mate connector at a rate of 25 ± 6 mm per minute. Per EIA-364-13.	0.50 N per Pin MAX Avg. mating force & 0.045 N per Pin MIN Avg. un-mating force
7	Durability	Mate connectors up to 100 cycles at a maximum rate of 10 cycles per minute. Per EIA-364-09.	10 milliohms MAXIMUM (change from initial)
8	Vibration (Random)	Mate connectors and vibrate 20-500Hz Random, 3.1g's. 15 minutes, 3 axes. Per EIA 364-28, test condition VII, condition D.	10 milliohms MAXIMUM (change from initial) & Discontinuity < 10 nanosecond
9	Shock (Mechanical)	Mate connectors and shock at 30 g's with $\frac{1}{2}$ sine wave (11 milliseconds) shocks in the $\pm X$, $\pm Y$, $\pm Z$ axes (18 shocks total). Per EIA-364-27, condition H.	10 milliohms MAXIMUM (change from initial) & Discontinuity < 10 nanosecond
10	Contact Normal Force	Deflect contacts at a rate of 25.4mm +/- 0.20mm per minute. Take normal force measurements at nominal deflection. Per EIA-364-04	0.2 N MINIMUM contact normal force per pin

5.3 ENVIRONMENTAL REQUIREMENTS

ITEM	DESCRIPTION	TEST CONDITION	REQUIREMENT
11	Thermal Shock	Mate connectors; expose to 10 cycles of: -55°C to 85°C. Per EIA-364-32, Test Condition 1.	10 milliohms MAXIMUM (change from initial) & Visual: No Damage
12	Temperature Life (Mated)	Mate connectors; expose to: 228 hours at $105 \pm 2^\circ\text{C}$. Per EIA-364-17	10 milliohms MAXIMUM (change from initial) & Visual: No Damage

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13	Cyclic Humidity (Mated)	Mate connectors: 10 cycles at temperature $25 \pm 3^{\circ}\text{C}$ at 80%-98% relative humidity and $65 \pm 3^{\circ}\text{C}$ at 90%-98% relative humidity. Per EIA-364-31, method III, omitting 7b vibration test. {Note: Remove surface moisture and air dry for 1 hour prior to measurements.}	10 milliohms MAXIMUM (change from initial) & Visual: No Damage
14	Thermal Disturbance	Cycle the connector between $15 \pm 3^{\circ}\text{C}$ and $85 \pm 3^{\circ}\text{C}$ as measured on the connector contacts. Ramps should be a average of 5°C per minute, and dwell times should insure that the contacts reach these temperature extremes (a minimum of 10 minutes). Humidity is not controlled. Perform 10 cycles on Mated connectors. Per EIA-364-110.	10 milliohms MAXIMUM (change from initial) & Visual: No Damage
15	Thermal Cycling	Cycle the connector between $15 \pm 3^{\circ}\text{C}$ and $85 \pm 3^{\circ}\text{C}$ as measured on the connector contacts. Ramps should be a average of 5°C per minute, and dwell times should insure that the contacts reach these temperature extremes (a minimum of 10 minutes). Humidity is not controlled. Perform 500 cycles on Mated connectors. Per EIA-364-110.	10 milliohms MAXIMUM (change from initial) & Visual: No Damage
16	Dust	Use benign dust. Only one half of the unmated connector shall be exposed to dust and then mated to unexposed half. Per EIA-364-91.	10 milliohms MAXIMUM (change from initial)
17	Mixed Flowing Gas (MFG)	10 days Un-mated, 4 days Mated. Per EIA-364-65, method 2A	10 milliohms MAXIMUM (change from initial) & Visual: No Damage

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1. TESTING REQUIREMENTS

6.1 TEST SEQUENCE PER EIA-364-1000

Sequence	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7
1	LLCR	LLCR	LLCR	LLCR	LLCR	LLCR	LLCR
2	Durability* (preconditioning)	Durability* (preconditioning)	Durability* (preconditioning)	Durability* (preconditioning)	Durability* (preconditioning)	Durability* (preconditioning)	DWV
3	LLCR	LLCR	LLCR	LLCR	LLCR	LLCR	Durability
4	Temperature Life	Thermal shock	Vibration	Temperature Life** (preconditioning)	Temperature Life** (preconditioning)	Dust	LLCR
5	LLCR	LLCR	LLCR	LLCR	LLCR	LLCR	DWV
6	Reseating (3 M/U cycles)	Cyclic Temperature & humidity	Shock	Mixed Flowing Gas	Thermal Cycling	Thermal Disturbance	-
7	LLCR	LLCR	LLCR	LLCR	LLCR	LLCR	-
8	-	Reseating (3 M/U Cycles)	-	Thermal Disturbance	Reseating (3 M/U cycles)	Reseating (3 M/U cycles)	-
9	-	LLCR	-	LLCR	LLCR	LLCR	-
10	-	-	-	Reseating (3 M/U cycles)	-	-	-
11	-	-	-	LLCR	-	-	-

Durability* preconditioning - perform 20 unplug/plug cycles
 Temperature Life** preconditioning – 37 hours @ 115degC, mated.

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